BUREAUCRACY

Every new information technology has its unexpected bottlenecks. It solves some old problems but creates new ones. In the early 1730s BCE, Narâmtani, a priestess in the Mesopotamian city of Sippar, wrote a letter (on a clay tablet) to a relative, asking him to send her a few clay tablets he kept in his house. She explained that her claim to an inheritance was being contested and she couldn't prove her case in court without those documents. She ended her message with a plea: 'Now, do not neglect me!'[16]

We don't know what happened next, but just imagine the situation if the relative searched his house but could not find the missing tablets. As people produced more and more documents, finding them turned out to be far from easy. This was a particular challenge for kings, priests, merchants and anyone else who accumulated thousands of documents in their archives. How do you find the right tax record, payment receipt or business contract when you need it? Written documents were much better than human brains in recording certain types of information. But they created a new and very thorny problem: retrieval.[17]

The brain is remarkably efficient in retrieving whatever information is stored in its network of tens of billions of neurons and trillions of synapses. Though our brain archives countless complex stories about our personal life, our national history and our religious mythology, healthy people can retrieve information about any of them in less than a second. What did you eat for breakfast? Who was your first crush? When did your country gain its independence? What's the first verse in the Bible?

How did you retrieve all these pieces of information? What mechanism activates the right neurons and synapses to rapidly call up the necessary information? Though neuroscientists have made some progress in the study of memory, nobody yet understands what memories are, or how exactly they are stored and retrieved.[18] What we do know is that millions of years of evolution streamlined the brain's retrieval processes. However, once humans outsourced memories from organic brains to inorganic documents, retrieval could no longer rely on that streamlined biological system. Nor could it rely on the foraging abilities that humans evolved over millions of years. Evolution has adapted humans for finding fruits and mushrooms in a forest, but not for finding documents in an archive.